

Paracetamol overdose: Texts

Text A

Paracetamol: contraindications and interactions

4.4 Special warnings and precautions for use

Where analgesics are used long-term (>3 months) with administration every two days or more frequently, headache may develop or increase. Headache induced by overuse of analgesics (MOH medication-overuse headache) should not be treated by dose increase. In such cases, the use of analgesics should be discontinued in consultation with the doctor.

Care is advised in the administration of paracetamol to patients with alcohol dependency, severe renal or severe hepatic impairment. Other contraindications are: shock and acute inflammation of liver due to hepatitis C virus. The hazards of overdose are greater in those with non-cirrhotic alcoholic liver disease.

4.5 Interaction with other medicinal products and other forms of interaction

- Anticoagulants – the effect of warfarin and other coumarins may be enhanced by prolonged regular use of paracetamol with increased risk of bleeding. Occasional doses have no significant effect.
- Metoclopramide – may increase speed of absorption of paracetamol.
- Domperidone – may increase speed of absorption of paracetamol.
- Colestyramine – may reduce absorption if given within one hour of paracetamol.
- Imatinib – restriction or avoidance of concomitant regular paracetamol use should be taken with imatinib.

A total of 169 drugs (1042 brand and generic names) are known to interact with paracetamol.

14 major drug interactions (e.g. amyl nitrite)

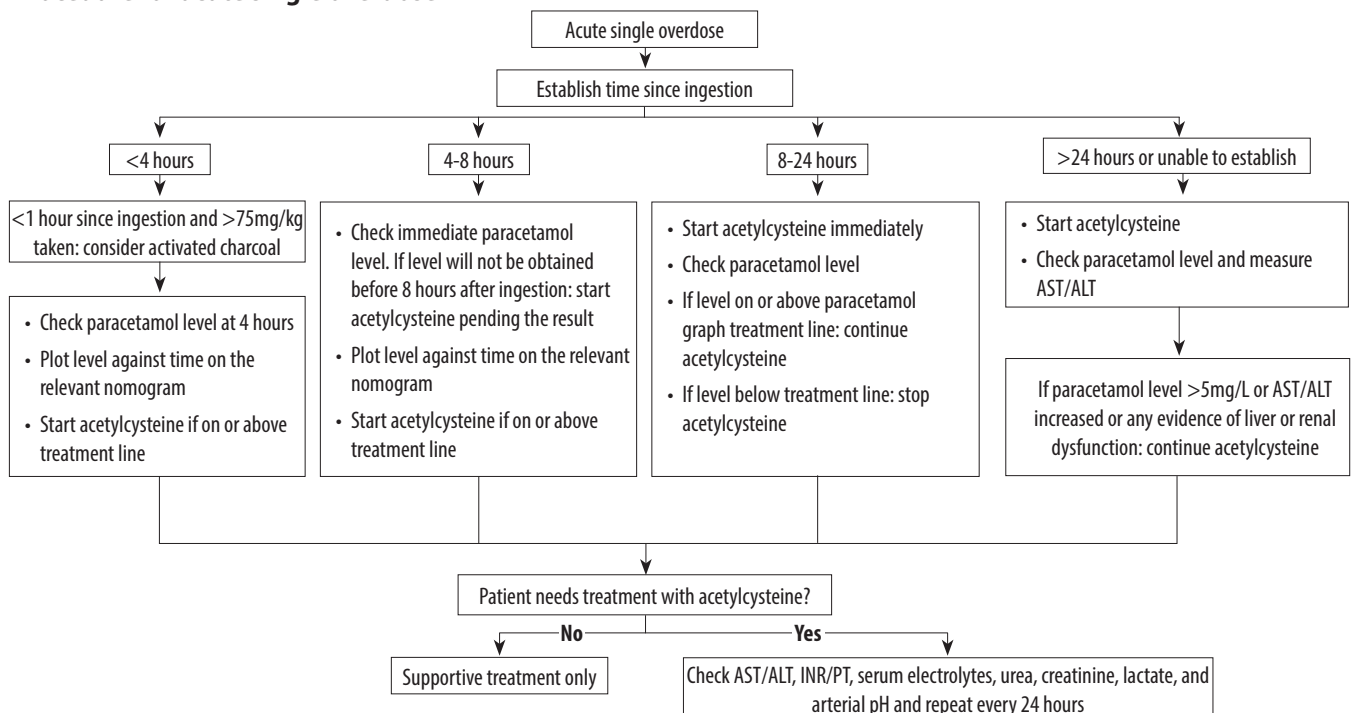
62 moderate drug interactions

93 minor drug interactions

A total of 118 brand names are known to have paracetamol in their formulation, e.g. Lemsip.

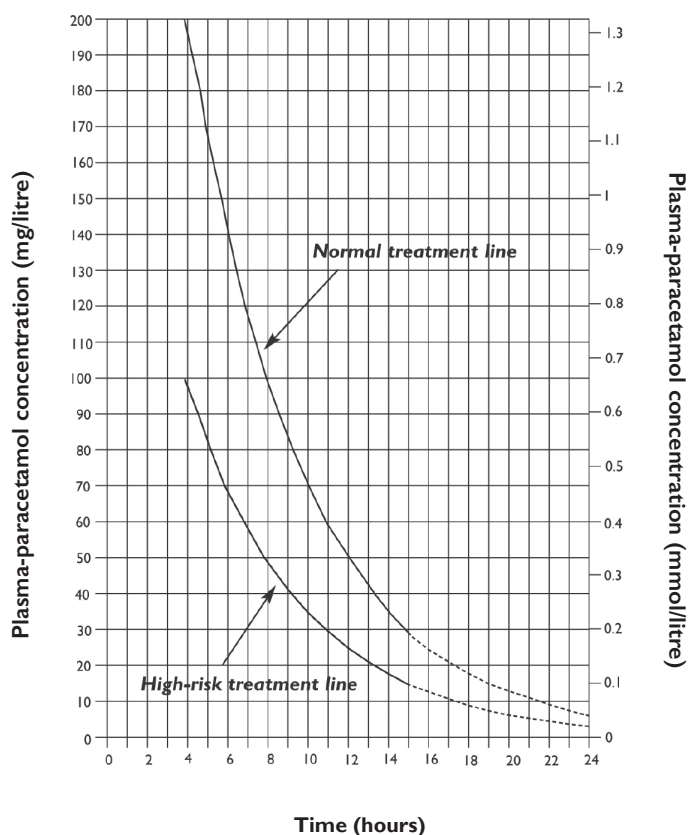
Text B

Procedure for acute single overdose



Text C

Paracetamol poisoning – Emergency treatment of poisoning



Patients whose plasma-paracetamol concentrations are above the **normal treatment line** should be treated with acetylcysteine by intravenous infusion (or, if acetylcysteine cannot be used, with methionine by mouth, provided the overdose has been taken **within 10-12 hours** and the patient is not vomiting).

Patients on enzyme-inducing drugs (e.g. carbamazepine, phenobarbital, phenytoin, primidone, rifampicin and St John's wort) or who are malnourished (e.g. in anorexia, in alcoholism, or those who are HIV positive) should be treated with acetylcysteine if their plasma-paracetamol concentration is above the **high-risk treatment line**.

Text D

Clinical Assessment

- Commonly, patients who have taken a paracetamol overdose are asymptomatic for the first 24 hours or just have nausea and vomiting
- Hepatic necrosis (elevated transaminases, right upper quadrant pain and jaundice) begins to develop after 24 hours and can progress to acute liver failure (ALF)
- Patients may also develop:
 - Encephalopathy
 - Renal failure – usually occurs around day three
 - Oliguria
 - Lactic acidosis
 - Hypoglycaemia

History

- Number of tablets, formulation, any concomitant tablets
- Time of overdose
- Suicide risk – was a note left?
- Any alcohol taken (acute alcohol ingestion will inhibit liver enzymes and may reduce the production of the toxin NAPQI, whereas chronic alcoholism may increase it)

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate **Text Booklet**.
- For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
- Write your answers on the spaces provided in this **Question Paper**.
- Answer all the questions within the 15-minute time limit.
- Your answers should **only** be taken from texts **A-D** and must be correctly spelt.

Paracetamol overdose: Questions

Questions 1-7

For each question, **1-7**, decide which text (**A**, **B**, **C** or **D**) the information comes from. Write the letter **A**, **B**, **C** or **D** in the space provided. You may use any letter more than once.

In which text can you find information about

- 1 the various symptoms of patients who have taken too much paracetamol? _____
- 2 the precise levels of paracetamol in the blood which require urgent intervention? _____
- 3 the steps to be taken when treating a paracetamol overdose patient? _____
- 4 whether paracetamol overdose was intentional? _____
- 5 the number of products containing paracetamol? _____
- 6 what to do if there are no details available about the time of the overdose? _____
- 7 dealing with paracetamol overdose patients who have not received adequate nutrition? _____

Questions 8-13

Answer each of the questions, **8-13**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both. You should **not** write full sentences.

- 8 If paracetamol is used as a long-term painkiller, what symptom may get worse?

- 9 It may be dangerous to administer paracetamol to a patient with which viral condition?

- 10 What condition may develop in an overdose patient who presents with jaundice?

11 What condition may develop on the third day after an overdose?

12 What drug can be administered orally within 10 - 12 hours as an alternative to acetylcysteine?

13 What treatment can be used if a single overdose has occurred less than an hour ago?

Questions 14-20

Complete each of the sentences, **14-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

14 If a patient has taken metoclopramide alongside paracetamol, this may affect the

_____ of the paracetamol.

15 After 24 hours, an overdose patient may present with pain in the _____.

16 For the first 24 hours after overdosing, patients may only have such symptoms as

_____.

17 Acetylcysteine should be administered to patients with a paracetamol level above the high-risk treatment line who are taking any type of _____ medication.

18 A non-high-risk patient should be treated for paracetamol poisoning if their paracetamol level is above _____ mg/litre 8 hours after overdosing.

19 A high-risk patient who overdosed _____ hours ago should be given acetylcysteine if their paracetamol level is 25 mg/litre or higher.

20 If a patient does not require further acetylcysteine, they should be given treatment categorised as _____ only.

END OF PART A

THIS QUESTION PAPER WILL BE COLLECTED